

Product Reference Manual (V1.0)

Description

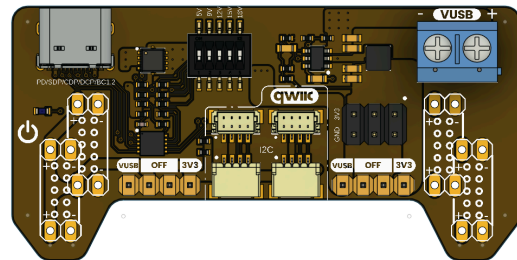
The DevLab Power Supply for Breadboard with Qwiic is a USB Type-C-powered power distribution module intended for breadboard-based electronic prototyping. The device operates as a USB Power Delivery (PD) sink and provides selectable output voltage rails for external circuits through standard breadboard headers, pin headers, and a screw terminal interface.

The module integrates a USB PD controller for voltage negotiation, a high-efficiency DC-DC regulator for 3.3 V generation, and configurable output routing through jumpers and DIP switches. Power outputs are distributed symmetrically on both sides of the board to support dual-rail breadboard layouts.

Electrical interfacing is provided through a QWIIC-compatible I²C connector, enabling optional external monitoring or control by a host microcontroller. All power and signal references share a common ground plane.

DevLab format compatibility.

Simplicity and compatibility are the core principles of the **DevLab form factor**. This standard defines a **compact and communication-optimized board layout**, ensuring straightforward connection and interoperability among DevLab modules.



Key Features

- USB Type-C power input with USB Power Delivery sink capability
- Support for multiple fixed output voltage levels: **3.3 V, 5 V, 9 V, 12 V, 15 V, 18 V, and 20 V**
- Configurable power routing for dual-rail breadboard layouts
- Integrated high-efficiency DC-DC regulator for 3.3 V output
- QWIIC-compatible I²C connector for optional external interfacing
- Breadboard-compatible mechanical footprint
- On-board power status indicator LED

Hardware Features

- USB Type-C connector for power input
- HUSB238 USB Power Delivery controller
- TPS54302 step-down regulator for 3.3 V rail generation
- DIP switch for output voltage selection
- Jumper-selectable output voltage routing for left and right breadboard rails
- Screw terminal output for VUSB power access
- Dual pin header outputs for breadboard power distribution
- Common ground reference across all outputs and connectors

Software Support

- **Not required**

Applications

- Breadboard-based electronic prototyping
- Power delivery evaluation for embedded systems
- Educational laboratories and training platforms
- Sensor and peripheral power supply development
- Early-stage hardware validation and testing

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1. The Board

1.1 Accessories

- 16x single-row male header
- 1x QWIIC 4-pin JST 1.0 mm Harness
- 2 x Jumper cap

2 Ratings

2.1 Recommended Operating Conditions

Parameter	Description	Min	Typ	Max	Unit
V_{in}	USB Type-C PD input voltage	5	-	20	V
V_{BUS}	USB PD negotiated output voltage	5	09/12/15	20	V
V_{3V3}	Regulated 3.3V output rail	3.2	3.3	3.4	V
I_{3V3}	Maximum 3.3V regulator output current	-	-	2	A
I_{BUS}	Maximum breadboard output current	-	-	5	A
V_{IH}	High-level input voltage for I ² C	0.7 × V _{CC}	-	5.5	V
V_{IL}	Low-level input voltage for I ² C	-	-	0.3 × V _{CC}	V
I_{CC}	Typical module supply current	-	3.1	-	mA
f_{I2C}	Supported I ² C communication frequency	0	100	400	kHz
TOPD	USB Power Delivery supported profiles	5	09 /12/15	20	V
T_{AMB}	Recommended operating ambient temperature	-20	25	85	

Note 1: USB Power Delivery voltage selection can be configured through the onboard DIP switch resistor network or dynamically through the HUSB238 I²C interface. I²C configuration has priority over DIP switch settings.

Note 2: The 3.3V regulator output capability depends on thermal dissipation, PCB airflow, and input voltage conditions.

Note 3: Qwiic/STEMMA QT connectors share the same I²C bus signals (3V3, GND, SDA, SCL) for daisy-chain operation.

Note 4: The maximum VBUS/PD output current depends on the power capability, voltage profile, cable quality, and electrical characteristics of the connected USB Type-C PD power adapter.

Note 5: The module integrates breadboard-compatible power rails with selectable VUSB and regulated 3.3V outputs.

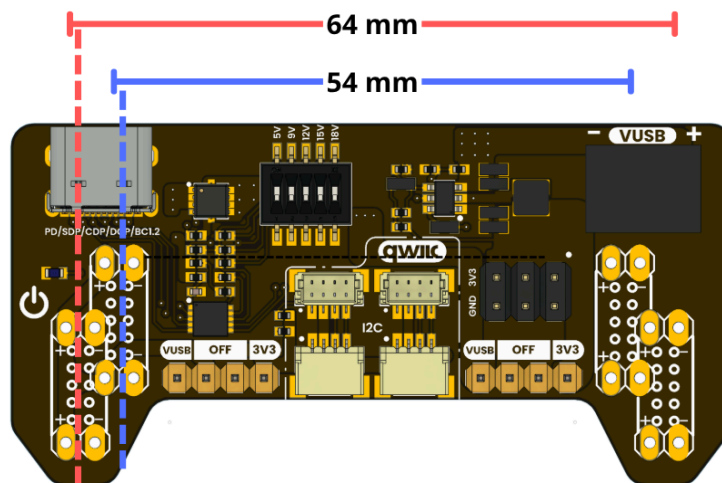
Note 6: Output voltages and currents may vary with the characteristics of the power supply

Note 7: For stable 3.3 V regulation, use **USB PD profiles of 9 V, 12 V, 15 V, or 20 V**. If operating from **5 V**, ensure the input remains above **4.5 V** under load; otherwise, the **TPS54302DDCT** may shut down due to **UVLO**.

3. Functional Overview

The DevLab Power Supply for Breadboard with I2C is an intelligent USB Type-C Power Delivery (PD) power module designed for rapid prototyping, embedded development, and laboratory integration. The board operates as a USB-C PD sink controller, automatically negotiating power profiles from a compatible USB-C charger and distributing the selected voltage to external systems while simultaneously maintaining a regulated 3.3 V rail for logic and control circuitry.

The module is mechanically compatible with standard 54 mm and 64 mm breadboards, allowing direct integration into prototyping environments without additional adapters or wiring complexity. Dedicated output rails distribute selectable VUSB and regulated 3.3 V lines to both sides of the breadboard, simplifying power distribution for mixed-voltage embedded systems.



At the center of the design, the HUSB238 controller manages USB-C Power Delivery communication. The device negotiates available voltage profiles directly with the connected PD source and enables selection of standard output voltages, including **5 V, 9 V, 12 V, 15 V, 18 V, and 20 V**. Voltage selection can be performed through the onboard DIP switch without requiring firmware, making the module suitable for standalone hardware operation and rapid hardware validation.

The negotiated VOUT rail is routed to the screw terminal outputs and breadboard power rails, enabling direct connection to high-power devices such as motors, LED systems, DC-DC converters, battery chargers, and embedded platforms requiring elevated supply voltages. In parallel, an onboard TPS54302 buck regulator generates a stable 3.3 V output for sensors, microcontrollers, and low-voltage digital peripherals.

For embedded connectivity and expansion, the board integrates multiple QWIIC-compatible JST 1 mm connectors, allowing direct interfacing with I²C sensors and modules. Optional solder jumpers permit direct connection of the HUSB238 I²C interface to the shared Qwiic bus, enabling firmware-based monitoring and dynamic voltage control from an external microcontroller.



The design also incorporates dedicated diagnostic and integration features, including PD configuration test points, USB signal probing pads, visual power indication through a status LED, and clearly separated top and bottom board layouts to simplify inspection, assembly, debugging, and validation during development.

3.1 Breadboard Compatibility and Mechanical Integration

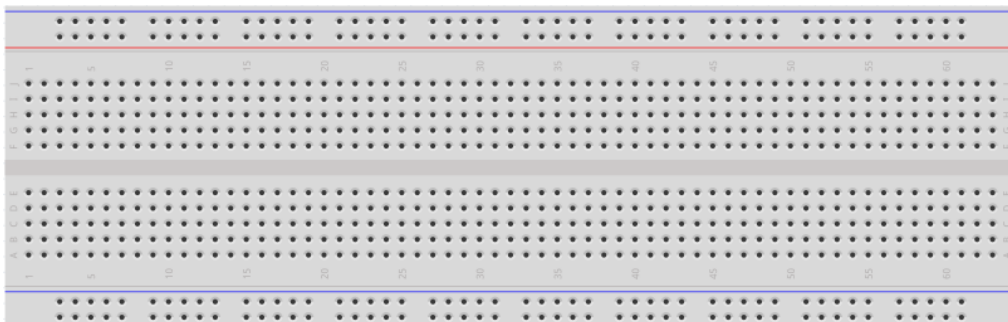
The power supply for breadboards with QWIIC is designed to mechanically integrate with standard solderless breadboards commonly used in embedded development, electronics education, and rapid prototyping environments. The PCB layout provides direct alignment with the power rails of both 54 mm and 64 mm breadboard formats, allowing the module to distribute power without requiring additional jumper wires or external adapters.

The board includes mirrored power output sections on both the left and right sides, enabling simultaneous power injection into dual breadboard rails. This configuration simplifies laboratory setups and improves cable organization during development and testing.

The mechanical spacing of the output headers was optimized to maintain compatibility with the most common breadboard dimensions while preserving sufficient clearance for USB-C connectors, QWIIC cables, and external terminal connections.

3.2 54 mm Breadboard Configuration

The 54 mm breadboard configuration is intended for compact prototyping systems and low-profile embedded developments. In this configuration, the module aligns directly with the narrower breadboard spacing while maintaining access to the USB-C connector, QWIIC interfaces, and screw terminal outputs



For 54 mm breadboards:

- The power rails align directly with the VOUT and GND outputs on both sides of the module.
- The selectable VUSB or 3.3 V outputs can independently power the breadboard rails.
- Compact mechanical spacing minimizes wiring complexity.
- Suitable for sensor evaluation, MCU development, and portable embedded systems.

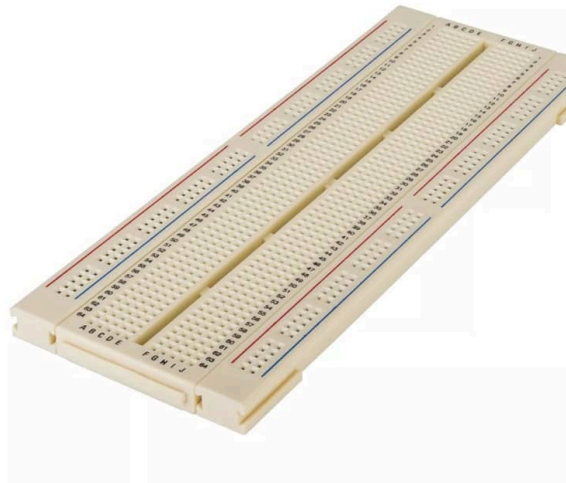
The onboard selector jumpers allow users to independently configure the left and right breadboard rails for the following:

- VUSB output
- OFF state
- Regulated 3.3 V output

This enables mixed-voltage prototyping where one side of the breadboard operates at higher PD voltages while the opposite side remains at safe 3.3 V logic levels.

3.3 64 mm Breadboard Configuration

The 64 mm breadboard configuration supports larger prototyping platforms with wider rail spacing and increased component density. The module maintains compatibility through extended output positioning and symmetric rail distribution.



For 64 mm breadboards:

- The output rail spacing matches standard wide-format breadboards.
- Independent rail voltage selection remains available on both sides.
- Additional clearance improves accessibility for large components and wiring.
- Suitable for power electronics, motor control evaluation, and mixed-signal systems.

This configuration is particularly useful for applications requiring the following:

- Multiple voltage domains
- High-current peripherals
- External DC-DC converters
- Distributed sensor networks
- Simultaneous analog and digital circuitry

The wider mechanical format also improves thermal distribution around the USB-C PD circuitry and power regulation stages during high-power operation.

3.4 Breadboard Power Rail Configuration

Each side of the module includes an independent voltage selection block that controls the voltage delivered to the breadboard power rails. The configuration is performed using onboard jumper selectors.

The available rail configurations are:

Selection	Function
VUSB	Routes negotiated USB-C PD voltage to the rail
OFF	Disconnects rail output
3V3	Routes regulated 3.3 V output to the rail

Because the left and right rails are independently configurable, the board supports asymmetric voltage distribution. For example:

Left Rail	Right Rail	Typical Application
3.3 V	3.3 V	Standard logic systems
VUSB	3.3 V	Mixed power and logic systems
12 V	3.3 V	Motor driver + MCU systems
OFF	VUSB	Single-rail high-voltage testing

This architecture increases flexibility during rapid prototyping and simplifies evaluation of systems operating across multiple voltage domains.

3.5 QWIIC Expansion and I²C Integration

The module integrates multiple QWIIC-compatible JST 1 mm connectors for simplified I²C expansion. These connectors expose the following:

- SDA
- SCL
- 3.3 V
- GND

The I²C bus operates exclusively at 3.3 V logic levels and is electrically isolated from the higher VUSB power rail.

Optional solder jumpers allow the HUSB238 controller to share the same I²C bus used by external QWIIC peripherals. This enables:

- Real-time PD monitoring
- Dynamic voltage selection
- Embedded telemetry
- Automated power negotiation

When the jumpers are left open, the HUSB238 remains electrically isolated from the external QWIIC network, preventing unintended interference in standalone operation.

3.6 Power Distribution Architecture

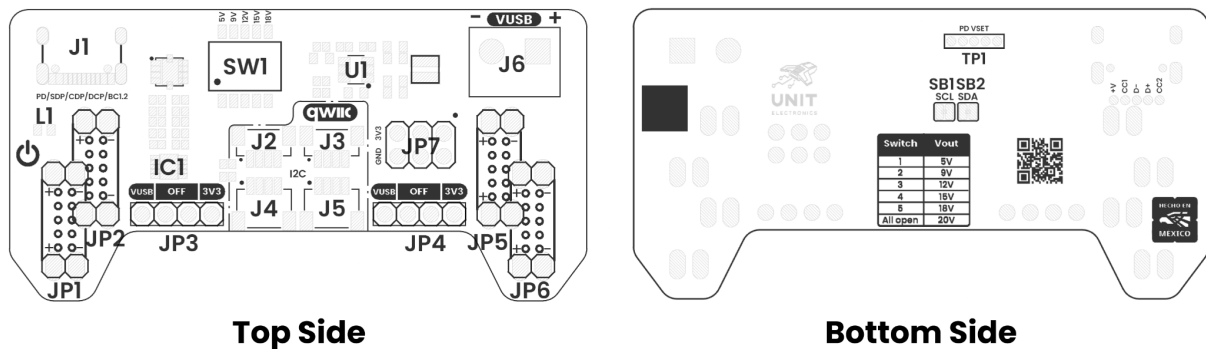
The board separates the high-voltage PD power domain from the low-voltage logic domain to improve stability and system safety.

The architecture consists of:

- USB-C PD input stage
- HUSB238 negotiation controller
- VUSB distribution rail
- TPS54302 buck converter
- Regulated 3.3 V rail
- Independent breadboard rail selectors

This separation allows high-voltage outputs such as 12 V, 15 V, or 20 V to coexist safely with 3.3 V digital systems on the same development platform.

3.7 Board Topology



Views of Board Topology

Views of Module Topology

Table 3.2.1 - Components Overview

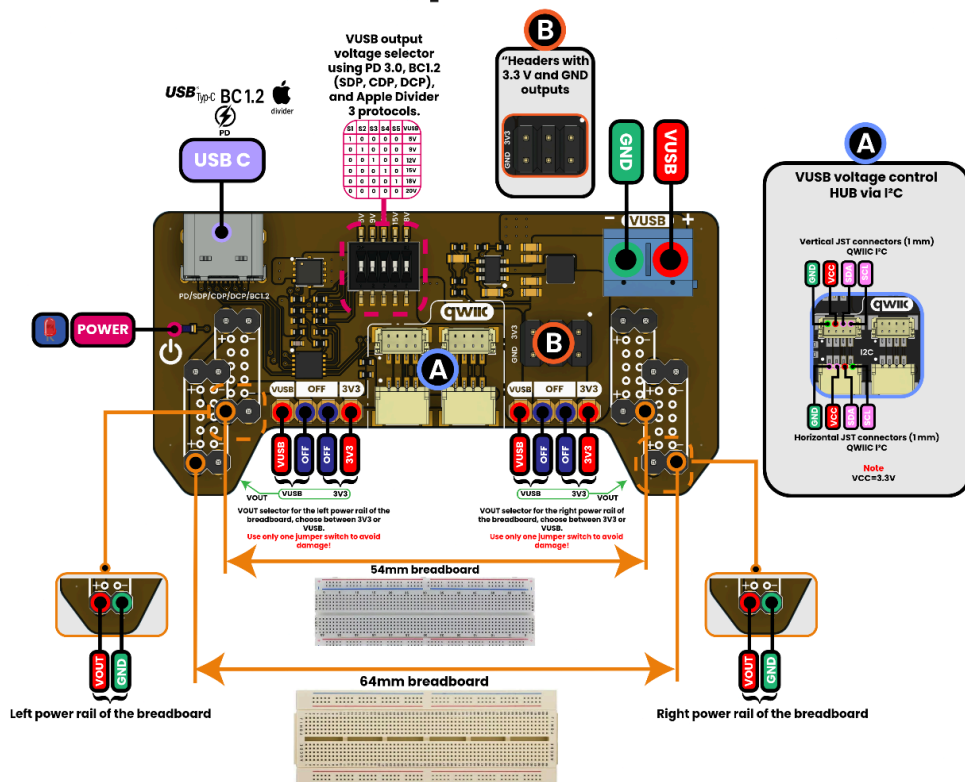
Ref.	Description
J1	USB Type-C Connector
L1	Power On LED
IC1	HUSB238
SW1	Dip Switch for voltage selector
U1	TPS54302 3.3V Regulator
J6	Screw Terminal Block output for VUSB
JP7	Pin Header for 3V3 power supply
JP3	Output voltage selector for the left side headers
JP4	Output voltage selector for the right side headers
JP1- JP2	Left side output voltage headers for breadboard
JP5- JP6	Right side output voltage headers for breadboard

4 General Pinout

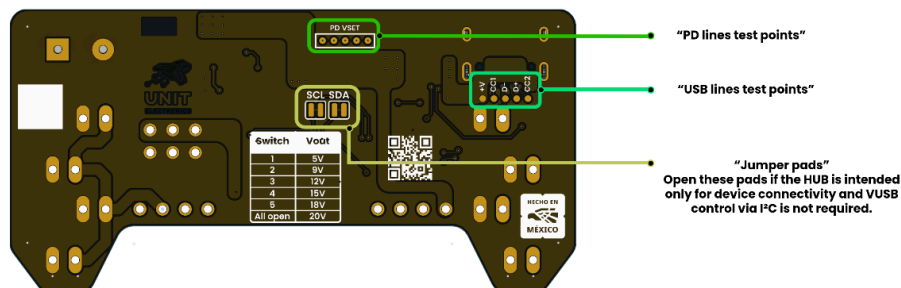
PINOUT

DevLab: Power Supply for Breadboard with Qwiic

Top view



Bottom view



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DevLab Module General Pinout

4.2 Pinout General Description

Pin Label	Function	Notes
VUSB	USB Power Delivery output rail	Selectable PD voltage output (5V, 9V, 12V, 15V, 20V)
3V3	Regulated 3.3V output	Generated by onboard TPS54302 regulator
GND	Ground reference	Common ground for power and signals
SDA	I ² C serial data line	Connected to Qwiic/STEMMA QT interface and HUSB238
SCL	I ² C serial clock line	Connected to Qwiic/STEMMA QT interface and HUSB238
USB-C	USB Type-C PD input connector	Power input supporting USB Power Delivery
DIP SW	PD voltage selection switch	Selects PD negotiation voltage profile
QWIIC	Qwiic/STEMMA QT connector	JST 1.0 mm connector for I ² C peripherals
VOUT	Breadboard power output	Output rail for external circuits and breadboards
GND OUT	Breadboard ground output	Ground rail for external circuits and breadboards
Header A	Breadboard interface header	Power rail connection for breadboard integration
Header B	Auxiliary power header	Provides VUSB, 3V3, and GND outputs

Note: The Qwiic/STEMMA QT connectors share the same SDA and SCL bus lines and are intended for daisy-chain I²C connectivity.

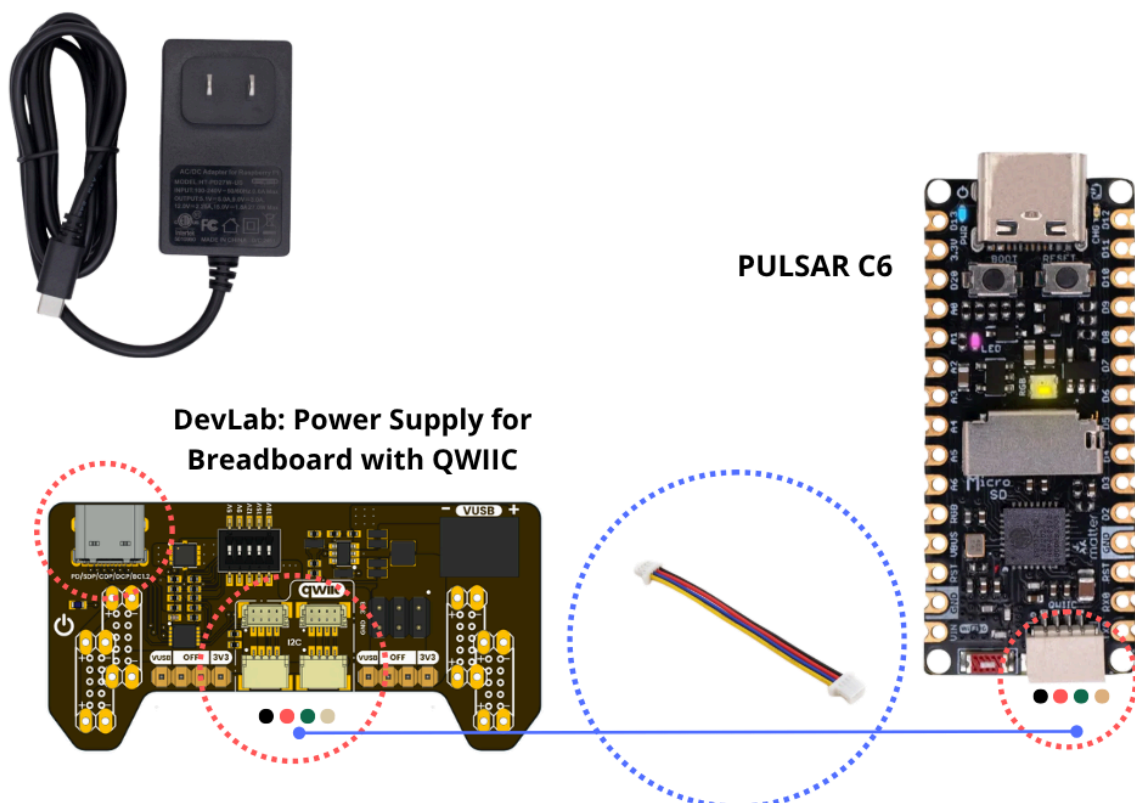
Note: Only one jumper configuration should be used at a time to avoid electrical conflicts or damage.

5 Board Operation

5.1 Getting Started with Arduino IDE

The DevLab Power Supply for Breadboard with Qwiic allows dynamic negotiation of USB Power Delivery (PD) voltages through an I²C interface. Using the integrated HUSB238 controller, embedded systems can request and monitor different voltage levels from a USB Type-C PD source, supporting output profiles from 5 V up to 20 V depending on charger capabilities.

This section demonstrates how to initialize the module, establish I²C communication, detect available PD voltage profiles, and verify the connection state using an ESP32-based platform such as the PULSAR C6.



Hardware Connections

Default DevLab / QWIIC Mapping

Power Supply	PULSAR C6 Pin	Description
VDD	3.3V	Power supply
GND	GND	Ground
SDA	GPIO6	I ² C Data

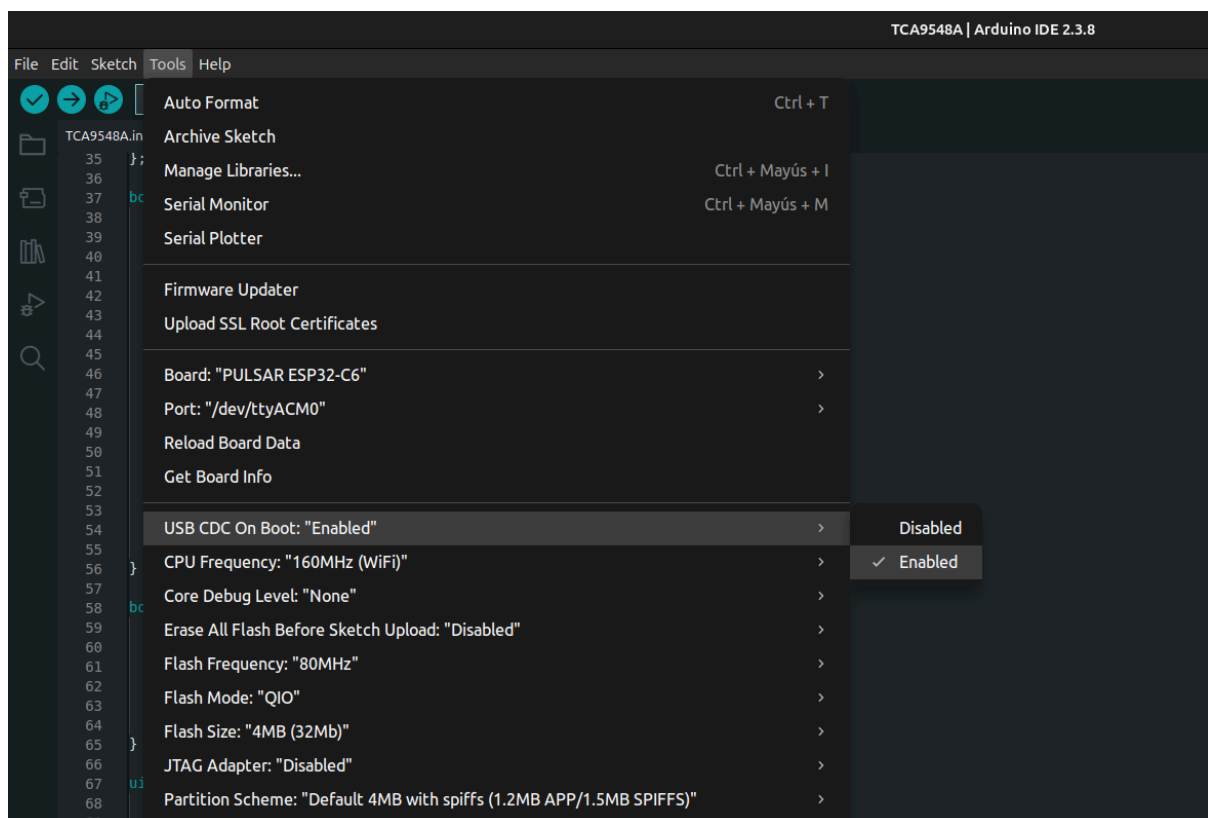
SCL	GPIO7	I ² C Clock
ADDR	GND	Default I ² C Address = 0x08

Important Electrical Notes

- The PULSAR C6 operates at **3.3 V logic**

5.3 Arduino IDE Configuration

- Open **Arduino IDE**
- Install ESP32 core:
 - Go to **Boards Manager**
 - Install: **ESP32 by Espressif Systems/UNIT Electronics ESP32**
- Select board:
 - ESP32C6 Dev Module/Pulsar C6
- 4. Select the correct serial port. For the Arduino example, enable USB CDC On Boot to use the serial monitor:



Required Libraries

Install the following library:

- Adafruit_HUSB238
- Wire (included in Arduino core)

Basic Initialization Example

The following example initializes the HUSB238 controller, verifies I²C communication, detects whether a USB-C PD source is attached, and displays the available voltage profiles.

```

/*
 * HUSB238 USB-C PD Module - Example 3: Detection Only
 *
 * Demo minima: solo verifica si el dispositivo USB-C esta conectado
 * y muestra los voltajes PD disponibles.
 */

#include <Wire.h>
#include "Adafruit_HUSB238.h"

#define I2C_SDA 6
#define I2C_SCL 7

Adafruit_HUSB238 husb238;

void printDetectedVoltages() {
  bool anyVoltage = false;

  Serial.println("Voltajes disponibles:");

  if (husb238.isVoltageDetected(PD_SRC_5V)) {
    Serial.println("5V");
    anyVoltage = true;
  }
  if (husb238.isVoltageDetected(PD_SRC_9V)) {
    Serial.println("9V");
    anyVoltage = true;
  }
  if (husb238.isVoltageDetected(PD_SRC_12V)) {
    Serial.println("12V");
    anyVoltage = true;
  }
  if (husb238.isVoltageDetected(PD_SRC_15V)) {
    Serial.println("15V");
    anyVoltage = true;
  }
  if (husb238.isVoltageDetected(PD_SRC_18V)) {
    Serial.println("18V");
    anyVoltage = true;
  }
}

```



```

}

if (husb238.isVoltageDetected(PD_SRC_20V)) {
    Serial.println("20V");
    anyVoltage = true;
}

if (!anyVoltage) {
    Serial.println("Ningun perfil PD detectado");
}
}

void setup() {
    Serial.begin(115200);
    while (!Serial) delay(10);

    Wire.begin(I2C_SDA, I2C_SCL);
    delay(1000);

    int maxRetries = 10;
    int retryCount = 0;
    bool initialized = false;

    while (retryCount < maxRetries && !initialized) {
        if (husb238.begin(HUSB238_I2CADDR_DEFAULT, &Wire)) {
            Serial.println("HUSB238 inicializado correctamente.");
            initialized = true;
        } else {
            retryCount++;
            if (retryCount < maxRetries) {
                Serial.print("Intento ");
                Serial.print(retryCount);
                Serial.println("fallido, reintentando...");
                delay(500);
            }
        }
    }

    if (!initialized) {
        Serial.println("ERROR: No se encontro HUSB238. Revisa el cableado I2C.");
    }
}

```

```
void loop() {
  if (!husb238.isAttached()) {
    Serial.println("Estado: DESCONECTADO");
    delay(2000);
    return;
  }

  Serial.println("Estado: CONECTADO");
  printDetectedVoltages();
  Serial.println("---");
  delay(3000);
}
```

Expected Serial Monitor Output

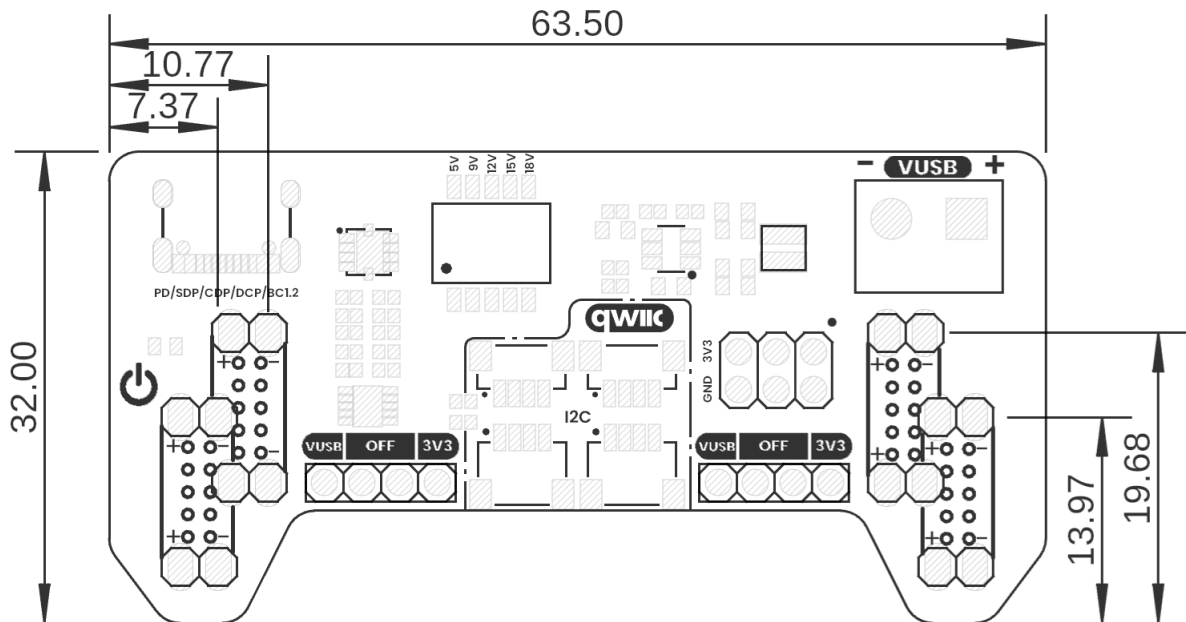
When the module is connected to a compatible USB-C PD charger, the serial monitor should display the detected voltage profiles available from the power source.

Example:

```
Status: CONNECTED
Available PD Voltages:
5V
9V
12V
15V
20V
---
```

If the module is disconnected or communication fails, the serial monitor will indicate the corresponding error or disconnected state.

6 Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	Link
Documentation	https://wiki.uelectronics.com/wiki/unit_devlab_power_supply_for_breadboard_with_qwiic
Github	https://github.com/UNIT-Electronics-MX/unit_devlab_power_supply_for_breadboard_with_qwiic
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1 Schematic

(https://github.com/UNIT-Electronics-MX/unit_devlab_power_supply_for_breadboard_with_qwiic/blob/main/hardware/unit_sch_v_1_0_0_ue0113_devlab_power_supply_for_breadboard_with_qwiic.pdf)

